

Day 2: Vector Space Model Summary

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Text data is symbolic, but computing platforms require continuous numerical representation to analyze it. Moreover, the same word can have different meanings based on its context



Solution:

Geometric Representation: words or complete documents are mapped out as mathematical vectors inside n dimensional mathematical space. The relative distance and alignment between vectors reflect semantic similarities or differences

VSM Structure Designs

Type

- Word by Word
- Word by Document

Construction

Matrix built from co occurrences counts between words. It captures semantic similarity directly

Matrix built from word frequencies in documents

Use Case

Better for word similarity tasks

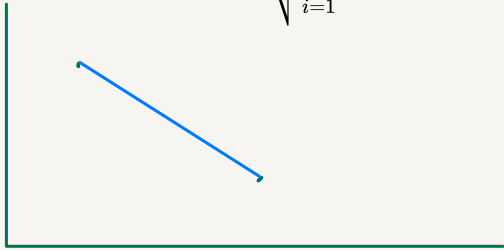
Useful for document classification and search

Similarly measures between vectors

- Euclidean Distance

- **Measures:** the absolute straight line distance between 2 points
- **Core Trait:** sensitive to scale of the vectors

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$



- **Interpretation :** the smaller the number, the more similar

- Cosine Similarity

- **Measures:** the angle between 2 vectors
- **Core Trait:** strongly preferred over distance

$$\text{cosine}(x, y) = \frac{x \cdot y}{\|x\| \|y\|}$$



- **Interpretation :**
 - -1 : Vectors are opposite
 - 1 : identical

Word Vectors

Type	Construction	Pros and cons
• Integer Representation	Each word gets an integer based on its position in the vocabulary	Little semantic sense (wrong or no ordering) Simple
• One Hot Vectors	Maps each unique word to an isolated vector containing a single 1 and filling all remaining by zeros	Very sparse, high dimensional (based on vocab size) and no embedded meanings Simple
• Dense Embeddings	Learns low dimensional vectors from the data itself	low dimensional vectors, and capture meaning and semantics

Simple Embedding Architectures

1 – Word2vec

Utilizes simple neural network to learn dense vector

- **Continuous Bag of Words** : Train the network to predict single center target word given its context
- **Skip Gram**: Takes a single center word and predicts its context words

Limitations: ignores word order within the context and produce same vector for all senses and different meanings of the word

2 – GloVe

Combines global matrix factorizations with local context windows. It directly constructs Embeddings by learning ratio of co occurrences counts across the complete corpus

3 – fastText

Expands word2vec by breaking individual vocab terms into character level n gram